Supplementary Material

Evaluating Tool-Using Language Agents: Judge Reliability, Propagation Cascades, and Runtime Mitigation in AgentProp-Bench

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1 Code, Data, and Trace Availability

All code, data, agent traces, and human labels are public at `https://github.com/bhaskargurram-ai/agenthallu-bench`. The repository contains: `data/results/` (result CSVs and analysis JSONs, including per-open-model baseline/P2 and ensemble re-judgements), `data/tasks/` (2,000 task specifications), `experiments/` (runners), `analysis/` (statistics, merge, tool-call-rate scripts), `metrics/` (EPS scorer, cascade detector), `agent/` (ReAct agent, model router with the local vLLM backend, tool executor), `benchmark/` (task generator and four domain simulators), `interceptor/`, and `tests/`. The command `bash analysis/finalize_openmodels.sh` regenerates every table and figure and recompiles the manuscript.

2 Open-Weight Model Serving and Tool-Call Adherence

Open-weight models were served locally with vLLM through an OpenAI-compatible endpoint and driven by the *same* prompted-ReAct harness as the proprietary models (the injection acts on the parsed tool call, which is model-agnostic). Table 1 reports the tool-call rate—the fraction of P2 semantic-wrong traces containing a parseable tool call—for every model evaluated. Only models that reliably emit tool calls yield measurable propagation; Mistral-7B-Instruct-v0.2 and Phi-3.5-mini did not and are therefore excluded from the propagation tables.

Table 1: Tool-call rate and fabricated-tool-use rate on P2 semantic-wrong traces. Fabricated tool use = no tool call emitted, yet the final answer asserts a completed tool-derived result.

Model	Type	Tool-call rate	Fabricated
GPT-4o	proprietary	100.0%	0.0%
GPT-4.1-nano	proprietary	99.0%	0.5%
o3-mini	proprietary	88.0%	8.0%
DeepSeek-V3	proprietary	86.0%	7.0%
GPT-4.1-mini	proprietary	85.0%	12.0%
Gemini-2.5-Flash	proprietary	76.0%	18.0%
GPT-3.5-Turbo	proprietary	59.5%	21.5%
GPT-4o-mini	proprietary	40.0%	24.0%
Gemini-2.0-Flash	proprietary	5.0%	42.5%
Qwen2.5-7B	open-weight	99.5%	—
Qwen2.5-3B	open-weight	70.0%	—
Qwen2.5-14B	open-weight	61.0%	—
Mistral-7B-v0.2	open-weight	21.5%	—
Phi-3.5-mini	open-weight	0.0%	—

3 Interceptor Threshold Sweep

Table 2 reports the 12-configuration sweep on the validation split (n=40 P2 rows). The reported F_1=0.842 is a sweep-internal optimum, not a held-out estimate.

Table 2: Full interceptor threshold sweep. Each row is a configuration of the three-layer filter: L1 (tool-call repetition threshold), L2 (keyword match count), L3 (LLM self-check). Best configuration by F_1 is highlighted. $N = 960$ P2 sessions.

L1 thresh	L2 kw	L3	F_1	Precision	Recall	Abstain%
1	2	✓	0.842 ⋆	0.923	0.774	65.0
1	2	—	0.809	0.826	0.792	57.5
1	3	✓	0.808	0.955	0.700	55.0
1	3	—	0.776	0.826	0.731	57.5
2	2	✓	0.762	0.842	0.696	47.5
2	2	—	0.780	0.889	0.696	45.0
2	3	✓	0.810	0.850	0.773	50.0
2	3	—	0.833	0.938	0.750	40.0
3	2	✓	0.769	0.938	0.652	40.0
3	2	—	0.750	0.857	0.667	35.0
3	3	✓	0.667	0.824	0.560	42.5
3	3	—	0.800	0.933	0.700	37.5

Table 3: Self-consistency analysis ($k = 3$ runs per task). Unique answers is the median count of distinct answers across repetitions. Hallucination rate is measured by heuristic match. Conditions: baseline (no injection) vs. P2 (semantic-wrong injection).

Model	Condition	n	Halluc.%	Median uniq. ans.	Abstain%
DeepSeek-V3	baseline	30	30.0	2.0	46.7
DeepSeek-V3	p2$_s$*emantic*	30	23.3	3.0	60.0
GPT-4o-mini	baseline	30	20.0	3.0	70.0
GPT-4o-mini	p2$_s$*emantic*	30	23.3	3.0	56.7

4 Self-Consistency Baseline (k=3)

Table 3 reports k=3 self-consistency voting on 30 tasks per condition; at n=30 this comparison is directional only.

5 Cross-Domain Replication: Retail

Table 4 reports baseline correctness and mean EPS on the 300 held-out retail tasks.

6 Interceptor Layer Ablation

Table 5 ablates the interceptor layers on GPT-4o-mini.

7 Power Analysis

Table 6 reports Cohen's h and achieved power for each statistical claim.

Table 4: Cross-domain transfer: held-out retail tasks. Models were not exposed to retail tasks during P2 tuning. Correctness judged by heuristic match. 95% bootstrap CIs ($B = 1{,}000$).

Model	n_{BL}	BL%	CI	n_{P2}	P2%	CI
DeepSeek-V3	100	37.0	[27.0, 47.0]	100	43.0	[33.0, 53.0]
GPT-4o-mini	100	29.0	[20.0, 38.0]	100	33.0	[24.0, 42.0]
Gemini-2.0-Flash	100	17.0	[10.0, 25.0]	100	17.0	[10.0, 25.0]

Table 5: Interceptor ablation study. Each condition removes one layer of the three-layer interceptor. Hallucination rate and abstain rate are reported. 95% bootstrap CIs ($B = 1{,}000$).

Model	Condition	n	Halluc.%	Abstain%	Correct%
Gemini-2.0-Flash	$\text{l1}_l\text{2}$	60	51.7	0.0	48.3
Gemini-2.0-Flash	$\text{l1}_l\text{2}_l\text{3}$	60	48.3	1.7	50.0
Gemini-2.0-Flash	$\text{l1}_o nly$	60	48.3	0.0	51.7
GPT-4o-mini	$\text{l1}_l\text{2}$	60	46.7	11.7	41.7
GPT-4o-mini	$\text{l1}_l\text{2}_l\text{3}$	60	41.7	26.7	31.7
GPT-4o-mini	$\text{l1}_o nly$	60	51.7	18.3	30.0

8 Per-Domain Propagation and Human Calibration

9 Ensemble Judge Prompt

The identical prompt used for all three LLM judges (GPT-4o, Gemini-2.5-Flash, GPT-4o-mini):

```
You are an impartial judge evaluating whether an agent's final answer is
   correct.
QUESTION: {query}
REFERENCE ANSWER: {reference}
AGENT ANSWER: {candidate}
Respond with a single token: YES if the agent answer substantively matches
    the
reference, NO otherwise. Accept minor wording differences, units expressed
differently, or extra explanation as long as the core fact is correct.
YOUR VERDICT (YES or NO):
```

10 Dataset Statistics

Table 6: Post-hoc power analysis for all main contrasts. Effect size is Cohen's h; achieved power is computed at $\alpha = 0.05$ (two-sided). Contrasts with power < 0.80 are flagged (†).

Contrast	n_1	n_2	p_1	p_2	h	Power
DeepSeek-V3: P2 vs BL	450	200	0.482	0.625	0.288	0.924
GPT-3.5-Turbo: P2 vs BL	450	200	0.496	0.675	0.366	0.991
GPT-4.1-mini: P2 vs BL	450	200	0.453	0.655	0.409	0.998
GPT-4.1-nano: P2 vs BL	450	200	0.547	0.845	0.668	1.000
GPT-4o-mini: P2 vs BL	450	200	0.600	0.710	0.232	0.780†
Gemini-2.0-Flash: P2 vs BL	450	200	0.524	0.670	0.298	0.939
Qwen2.5-14B: P2 vs BL	450	200	0.444	0.625	0.364	0.990
Qwen2.5-3B: P2 vs BL	450	200	0.458	0.825	0.792	1.000
Qwen2.5-7B: P2 vs BL	450	200	0.509	0.715	0.427	0.999
o3-mini: P2 vs BL	50	50	0.600	0.700	0.210	0.182†
GPT-4o-mini: Interceptor	600	600	0.558	0.328	-0.467	1.000
Gemini-2.0-Flash: Interceptor	600	600	0.445	0.432	-0.027	0.068†

Table 7: Dataset and trace statistics. P2 counts are across all four injection types; the propagation analysis uses the semantic-wrong subset.

Domain	Tasks	Baseline traces	Role
Calendar	667	per model	train/eval
Weather	674	per model	train/eval
Medical	559	per model	train/eval
Knowledge	100	per model	ablation (no injection)
Retail (held-out)	300	300	cross-domain replication

Models: 9 proprietary + 3 open-weight (Qwen2.5 3B/7B/14B).
Baseline traces: 4,100. **P2 traces (all types):** 9,400.
Propagation analysis (semantic-wrong): 1,950 across 12 models.
Human labels: 100. **Released agent traces:** >13,000.

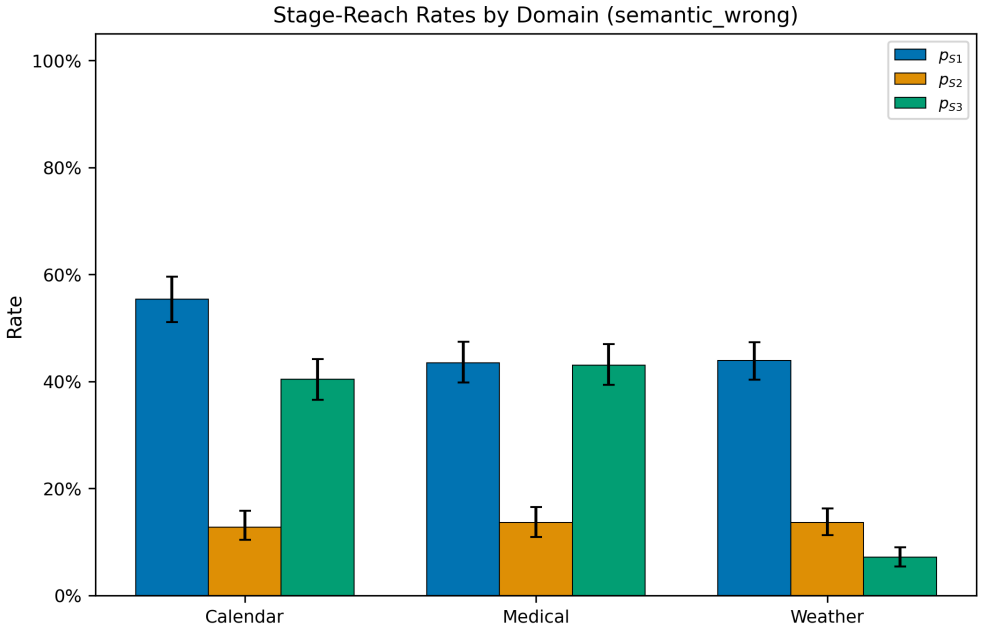


Figure 1: Per-domain S_1, S_2, S_3 rates under P2 semantic-wrong injection (ensemble-judged).

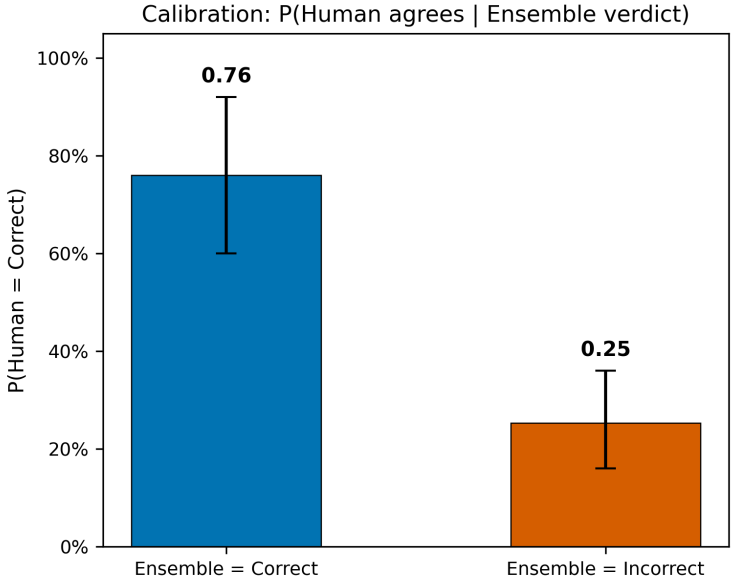


Figure 2: Calibration of the ensemble judge against 100 human labels: P(human=correct | ensemble=correct)=0.76; P(human=correct | ensemble=wrong)=0.25.